The Quantum Geometric Origin of Capacitance in Insulators

I. Komissarov, T. Holder & R. Queiroz

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Chen-Chih Wang

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Content

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Dielectric Response, Quantum Geometry, and Bounds on Optical Gaps

1. **The quantum geometric origin of capacitance in insulators**
   Authors: Ilia Komissarov, Tobias Holder, and Raquel Queiroz
   Nat. Commun. **15**, 4621 (2024); https://doi.org/10.1038/s41467-024-48808-x
   https://doi.org/10.48550/arXiv.2306.08035

2. **Universal relation between energy gap and dielectric constant**
   Authors: Yugo Onishi and Liang Fu
   https://doi.org/10.48550/arXiv.2401.04180

Recommended with a Commentary by Steven M. Girvin (Yale Quantum Institute) & Kun Yang (Florida State University)

Introduction and General Ideas

Quantum Geometry

Quantum geometry reflects the geometry of the Hilbert space. Here we use **momentum** as the continuous quantum number, and consider only **one band**

$$\mathrm{d}s^2 \equiv \| \, |u_n(\mathbf{k}+\mathrm{d}\mathbf{k})\rangle - |u_n(\mathbf{k})\rangle \, \|^2 = \langle \partial_{k_\mu} u_n | \partial_{k_\nu} u_n \rangle \mathrm{d}k^\mu \mathrm{d}k^\nu \equiv Q_{\mu\nu} \mathrm{d}k^\mu \mathrm{d}k^\nu$$

We further impose the covariant constraint

$$Q_{\mu\nu} = \langle \partial_{k_\mu} u_n | \partial_{k_\nu} u_n \rangle \to \langle D_{k_\mu} u_n | D_{k_\nu} u_n \rangle \quad ; \quad D_{k_\mu} \equiv \partial_{k_\mu} + iA_\mu, \quad A_\mu \equiv i \langle u_n | \partial_{k_\mu} u_n \rangle$$

Define $\mathcal{P} := |u_n\rangle \langle u_n|$ as the projector,

$$Q_{\mu\nu} = \left\langle \partial_{k_\mu} u_n \right| (\mathbb{1} - \mathcal{P}) \left| \partial_{k_\nu} u_n \right\rangle$$

Such a tensor can be separated into real and imaginary parts,

$$Q_{\mu\nu} \equiv g_{\mu\nu} + \frac{i}{2} \Omega_{\mu\nu}$$

Here $g_{\mu\nu}$ is called *quantum metric* and $\Omega_{\mu\nu}$ is exactly the *Berry curvature*.

R. Resta, Eur.Phys.J.B **79**, 121 (2011)

Quantum Geometry

Berry curvature:

$$\Omega_{\mu\nu} = D_{k_\mu} A_\nu - D_{k_\nu} A_\mu = i \left(\langle D_{k_\mu} u_n | D_{k_\nu} u_n \rangle - \langle D_{k_\nu} u_n | D_{k_\mu} u_n \rangle \right) = 2\mathrm{Im} \left\{ Q_{\mu\nu} \right\}$$

And the quantum metric is the real part,

$$g_{\mu\nu} = \frac{Q_{\mu\nu} + Q_{\nu\mu}}{2} = \langle u_n | \, r_\mu r_\nu \, | u_n \rangle - \langle u_n | \, r_\mu \, | u_n \rangle \langle u_n | \, r_\nu \, | u_n \rangle$$

If we now consider two bands, $\mathbb{1} = | u_n \rangle \langle u_n | + | u_m \rangle \langle u_m |$, then the formula becomes

$$g^{\mu\nu}_{nm} = \langle u_n | \, r_\mu \, | u_m \rangle \langle u_m | \, r_\nu \, | u_n \rangle \equiv r^\mu_{nm} r^\nu_{mn} = \frac{1}{2} \left(r^\mu_{nm} r^\nu_{mn} + r^\nu_{nm} r^\mu_{mn} \right)$$

With this, we can express both g and Ω in multiband cases as the following

$$\begin{cases} 2g^{\mu\nu}_{nm} = r^\mu_{nm} r^\nu_{mn} + r^\nu_{nm} r^\mu_{mn} \\ -i\Omega^{\mu\nu}_{nm} = r^\mu_{nm} r^\nu_{mn} - r^\nu_{nm} r^\mu_{mn} \end{cases}$$

Trace condition:

$$\det Q = \det g - \frac{\Omega^2}{4} \geq 0$$

$$\mathrm{tr} g \geq 2\sqrt{\det g} \geq |\Omega|$$

R. Resta, Eur.Phys.J.B **79**, 121 (2011)

Quantum Metric and Wannier Function

Quantum metric reflects the size of the Wannier function.

$$|u_n(\mathbf{k})\rangle = \int \mathrm{d}\mathbf{r} W_n(\mathbf{r}) e^{-i\mathbf{k}\cdot\mathbf{r}}$$

Thus

$$|u_n(\mathbf{k}+\mathrm{d}\mathbf{k})\rangle - |u_n(\mathbf{k})\rangle = \mathrm{d}\mathbf{k}\cdot\boldsymbol{\nabla}|u_n(\mathbf{k})\rangle = -i\mathrm{d}\mathbf{k}\cdot\int \mathrm{d}\mathbf{r} W_n(\mathbf{r})\mathbf{r} e^{-i\mathbf{k}\cdot\mathbf{r}}$$

So the infinitesimal distance is

$$\mathrm{d}s^2 = \underbrace{\left(\int \mathrm{d}\mathbf{r} W_n^*(\mathbf{r}) r^\mu e^{i\mathbf{k}\cdot\mathbf{r}}\right)\left(\int \mathrm{d}\mathbf{r} W_n(\mathbf{r}) r^\nu e^{-i\mathbf{k}\cdot\mathbf{r}}\right)}_{g_{\mu\nu}(\mathbf{k})} \mathrm{d}k^\mu \mathrm{d}k^\nu$$

Therefore, the integral of quantum metric reads

$$\int_{\mathrm{BZ}} \mathrm{d}\mathbf{k} g_{\mu\nu}(\mathbf{k}) = \int \mathrm{d}\mathbf{r}\,|W_n(\mathbf{r})|^2 r^\mu r^\nu \equiv \langle r^\mu r^\nu\rangle$$

Conductivity and Capacitance

Consider $j^\mu = \mathrm{d}P^\mu/\mathrm{d}t$

$$\sigma^{\mu\nu}(\omega) = \frac{\delta j^\nu}{\delta E^\mu} = \frac{\delta}{\delta E^\mu}\left(\frac{\mathrm{d}P^\nu}{\mathrm{d}t}\right)\bigg|_{E^\mu=0}$$

On the other hand, by linear response approach,

$$\begin{aligned}\sigma^{\mu\nu}(\omega) &= -\frac{e^2}{h}C\epsilon^{\mu\nu} + i\omega 4\pi\chi\delta^{\mu\nu} \\ &\equiv -\frac{e^2}{h}C\epsilon^{\mu\nu} + i\omega c\delta^{\mu\nu}\end{aligned}$$

Optical conductivity:

$$\varepsilon(\omega) = 1 + \frac{4\pi i\sigma(\omega)}{\omega} = 1 + 4\pi\chi(\omega)$$

Motivation:

- Does the intrinsic capacitance c depend on the quantum geometry?
- Is c a substantial contribution in quantum materials?

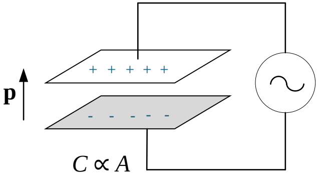
Conductivity and Capacitance

- P^μ in an insulator is determined entirely by the eigenfunctions and is related to the **Berry phase**

$$P^\mu = -e\left\langle r^\mu\right\rangle = -e\int_{\mathrm{BZ}} \mathrm{d}\mathbf{k}\sum_n f_n(\mathbf{k})\left\langle u_n(\mathbf{k})\right|\hat{r}^\mu\left|u_n(\mathbf{k})\right\rangle = -i\hbar e\int_{\mathrm{BZ}} \mathrm{d}\mathbf{k}\sum_n f_n(\mathbf{k})A_n(\mathbf{k})$$

(Note: This is the zeroth-order approximation. Later, one has to include the interaction with light.)

- The definition $\frac{\delta}{\delta E^\mu}\left(\frac{\mathrm{d}P^\nu}{\mathrm{d}t}\right)$ plus the Berry phase expression shows the geometric nature of the theory.
- Expressing the conductivity through polarization shows the direct relation between the **polarizability** and the **intrinsic capacitor**.



Methods and Results

Result:

$$\begin{cases} \sigma^{\mu\nu}_{\text{inter}}(\omega) = -\dfrac{ie^2}{\hbar} \displaystyle\sum_{n\neq m} \int_{\text{BZ}} \mathrm{d}\mathbf{k} f_{nm}\omega_{nm} \frac{r^{\mu}_{nm} r^{\nu}_{mn}}{\omega_{nm} + \omega} \\ \sigma^{\mu\nu}_{\text{intra}}(\omega) = \dfrac{i}{\omega} \displaystyle\sum_{n} \int_{\text{BZ}} f'_n J^{\mu}_{nn} J^{\nu}_{nn} \implies \text{vanishes in insulators} \end{cases}$$

Consider quantum metric

$$\begin{cases} 2g^{\mu\nu}_{nm} = r^{\mu}_{nm} r^{\nu}_{mn} + r^{\nu}_{nm} r^{\mu}_{mn} \\ -i\Omega^{\mu\nu}_{nm} = r^{\mu}_{nm} r^{\nu}_{mn} - r^{\nu}_{nm} r^{\mu}_{mn} \end{cases}$$

Eventually,

$$\begin{aligned} \sigma^{\mu\nu}_{\text{inter}}(\omega) = i\omega \frac{2e^2}{\hbar} \sum_{n\neq m} \int_{\text{BZ}} \mathrm{d}\mathbf{k} f_n (1 - f_m) \frac{\omega_{mn}}{\omega^2_{mn} - \omega^2} g^{\mu\nu}_{nm} \\ - \frac{e^2}{\hbar} \sum_{n\neq m} \int_{\text{BZ}} \mathrm{d}\mathbf{k} f_n (1 - f_m) \frac{\omega^2_{mn}}{\omega^2_{mn} - \omega^2} \Omega^{\mu\nu}_{nm} \end{aligned}$$

Capacitance and Quantum Metric

In the DC limit ($\omega \to 0$),

$$\sigma^{\mu\nu}_{\text{inter}}(\omega) = -\frac{e^2}{\hbar}\underbrace{\sum_{n\neq m}\int_{\text{BZ}} \mathrm{d}\mathbf{k} f_n(1-f_m)\Omega^{\mu\nu}_{nm}}_{\text{Chern number} \quad C\epsilon^{\mu\nu}}$$

$$+ i\omega \underbrace{\frac{2e^2}{\hbar}\sum_{n\neq m}\int_{\text{BZ}} \mathrm{d}\mathbf{k} f_n(1-f_m)\frac{1}{\omega_{mn}} g^{\mu\nu}_{nm}}_{\text{Intrinsic capacitance} \quad c^{\mu\nu}}$$

Then we get an important result,

$$c^{\mu\mu} = \frac{2e^2}{\hbar}\sum_{n\neq m}\frac{1}{\omega_{mn}}\int_{\text{BZ}} \mathrm{d}\mathbf{k} f_n(1-f_m) g^{\mu\nu}_{nm} \tag{1}$$

which bridges the **intrinsic capacitance** with the **quantum metric**.

Capacitance and Quantum Metric

Short Summary:

- **Multiband nature** is crucial since the intraband contribution is zero.
- The physical picture of the fact $\chi \propto c \propto \int_{\mathrm{BZ}} g^{\mu\nu}$ is that the more extended the wave pocket, the larger the polarizability.
- There is a rigorous relation between the **dielectric constant** and the **quantum metric** $\Longrightarrow$ We can use optical properties of an insulator to estimate the quantum metric.

Examples: Gapped Dirac Hamiltonians

2D gapped Dirac Hamiltonian

$$H = k_\mu \sigma^\mu + \frac{\Delta}{2}\sigma^z$$

Use equation (1),

$$c^{\mu\mu} = \frac{2e^2}{\hbar} \sum_{n \neq m} \frac{1}{\omega_{mn}} \int_{\mathrm{BZ}} \mathrm{d}\mathbf{k} f_n (1 - f_m) g_{nm}^{\mu\nu}$$

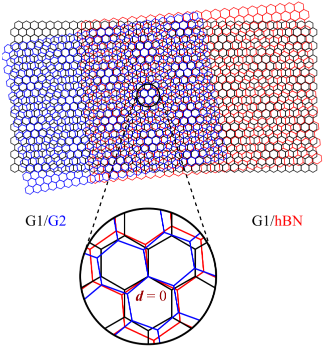
we have

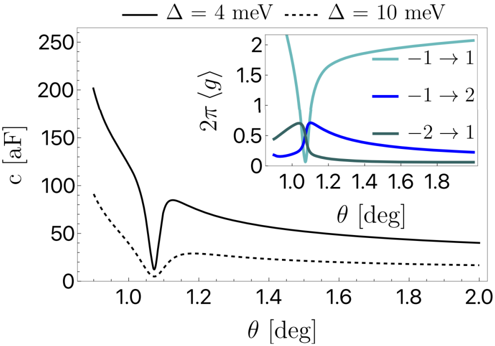
$$c = \mathrm{tr} c^{\mu\nu} \approx \frac{2e^2}{\Delta} \int_{\mathrm{BZ}} \mathrm{d}\mathbf{k}\ \mathrm{tr} g \geq \frac{e^2 C}{\pi \Delta}$$

This inequality tells us that the intrinsic capacitor diverges at a gap closing. In this example, we can see that the **polarizability** is related to the **band gap**.

Examples: Magic Angle Twisted Bilayer Graphene

TBG aligned on top of hBN. The in-plane inversion symmetry breaking leads to a gap Δ.





Trace condition $\mathrm{tr}g \geq |\Omega| / 2$ is saturated at the magic angle (LLL).

Dielectric Constant and Quantum Metric

From equation (1), the longitudinal dielectric constant is

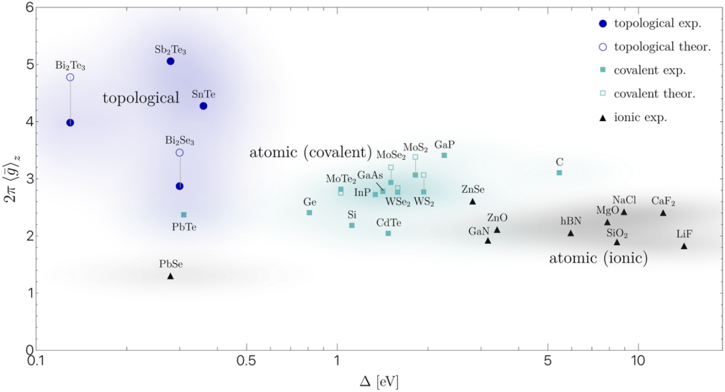
$$\varepsilon^{\mu\mu} = 1 + \frac{8\pi e^2}{\hbar} \sum_{n\neq m} \int_{\mathrm{BZ}} \mathrm{d}\mathbf{k} f_n(1-f_n) \frac{g_{mn}^{\mu\mu}}{\omega_{mn}} \leq 1 + \frac{8\pi e^2}{\Delta} \int_{\mathrm{BZ}} \mathrm{d}\mathbf{k} \underbrace{\sum_{n\neq m} f_n(1-f_n) g_{mn}^{\mu\mu}}_{g^{\mu\mu}}$$

This is entirely contributed by electric properties.
Subsequently,

$$\int_{\mathrm{BZ}} \mathrm{d}\mathbf{k}\ \mathrm{tr} g \geq \frac{\Delta}{8\pi e^2}(\varepsilon - 1)$$

In this perspective, we can estimate the quantum metric from the experimental values of **dielectric constant** ε and **band gap** Δ.

Dielectric Constant and Quantum Metric



$$\langle g\rangle_z \equiv \int \frac{a_z \mathrm{d}k_z}{2\pi}\int \frac{\mathrm{d}^2\mathbf{k}}{(2\pi)^2}\left(g^{xx}+g^{yy}\right) \geq \frac{4\pi}{e^2} a_z \Delta\left(\varepsilon-1\right) \equiv \langle\bar{g}\rangle_z$$

Conclusion

- The relation between conductivity in insulators and the quantum metric is established.
- In insulators, both the **polarizability** and the **conductivity** are completely determined by the properties of the Hilbert space (quantum geometry)
- Optical properties such as ε could be used to estimate the quantum metric.

Outlook

Quantum Weight (Y.Onishi & L.Fu, arXiv:2406.06783v3) The quantum metric is intimately related to the structure factor (number fluctuation)

$$S_{\mathbf{q}} \equiv \frac{1}{V} \langle \hat{n}_{\mathbf{q}} \hat{n}_{-\mathbf{q}} \rangle \approx \frac{1}{2\pi} K_{\alpha\beta} q^{\alpha} q^{\beta}$$

where $K_{\alpha\beta}$ is called *Quantum Weight*. It has a relation

$$K_{\mu\nu} = 2\pi \int_{\mathrm{BZ}} \frac{\mathrm{d}^D \mathbf{k}}{(2\pi)^D} g^{\mu\nu}(\mathbf{k})$$

(Which is related to $\chi \propto \langle \rho(r)\rho(r') \rangle$ from the perspecive provided by Fluctuation Dissipation Theorem)
At the same time, the entanglement entropy also has things to do with number correlation (G.C.Levine *et al.* PRA **83**, 013623 (2011))

$$\int \mathrm{d}r \mathrm{d}r' \langle n(r) n(r') \rangle \propto S$$

So it is promising to build a connection between **quantum entanglement** and **quantum geometry**.